

APS Mock Exam Solutions:

Q1) SALE! SALE! SALE!

Exercises:

- 4 → 16
- 5 → 20
- 17 → 70

Code:

```
num_items = int(input())

if num_items == 0:
    discount = 0
elif 1 <= num_items < 5:
    discount = 10 + (num_items-1)*2
elif 5 <= num_items < 10:
    discount = 20 + (num_items-5)*5
elif 10 <= num_items:
    discount = min(50 + (num_items-10)*10, 70)

print(discount)
```

Q2) Money Changer

Exercises:

- 7813
 - 5000 : 1
 - 1000 : 2
 - 500 : 1
 - 100 : 3
 - 50 : 0
 - 20 : 0
 - 10 : 1
 - 5 : 0
 - 2 : 1
 - 1 : 1
- 570
 - 5000 : 0
 - 1000 : 0
 - 500 : 1
 - 100 : 0
 - 50 : 1
 - 20 : 1
 - 10 : 0
 - 5 : 0

- 2:0
- 1:0
- 9
 - 5000:0
 - 1000:0
 - 500:0
 - 100:0
 - 50:0
 - 20:0
 - 10:0
 - 5:1
 - 2:2
 - 1:0

```

amount = 10200
fivek = 0
onek = 0
fiveh = 0
oneh = 0
fifty = 0
twenty = 0
ten = 0
five = 0
two = 0
one = 0

while amount > 0:
    if amount >= 5000:
        fivek = amount//5000
        amount = amount % 5000
    elif amount >= 1000:
        onek = amount//1000
        amount = amount % 1000
    elif amount >= 500:
        fiveh = amount//500
        amount = amount % 500
    elif amount >= 100:
        oneh = amount//100
        amount = amount % 100
    elif amount >= 50:
        fifty = amount//50
        amount = amount % 50
    elif amount >= 20:
        twenty = amount//20
        amount = amount % 20
    elif amount >= 10:
        ten = amount//10
        amount = amount % 10
    elif amount >= 5:

```

```

        five = amount//5
        amount = amount % 5
    elif amount >= 2:
        two = amount//2
        amount = amount % 2
    else:
        one = amount

print(5000, ":", fivek)
print(1000, ":", onek)
print(500, ":", fiveh)
print(100, ":", oneh)
print(50, ":", fifty)
print(20, ":", twenty)
print(10, ":", ten)
print(5, ":", five)
print(2, ":", two)
print(1, ":", one)

```

Q3) Patterns using Nested Iterations

Exercises:

- 3
 - 1
 - 2 1
 - 4 2 1
- 10
 - 1
 - 2 1
 - 4 2 1
 - 8 4 2 1
 - 16 8 4 2 1
 - 32 16 8 4 2 1
 - 64 32 16 8 4 2 1
 - 128 64 32 16 8 4 2 1
 - 256 128 64 32 16 8 4 2 1
 - 512 256 128 64 32 16 8 4 2 1
- 7
 - 1
 - 2 1
 - 4 2 1
 - 8 4 2 1
 - 16 8 4 2 1
 - 32 16 8 4 2 1
 - 64 32 16 8 4 2 1

Code:

```
for i in range(n):
    for j in range(i, -1, -1):
        print(2**j, end=" ")
    print()
```

Q4) Candy Bag

Exercises:

- $40 \rightarrow 0$
- $154 \rightarrow 2$
- $101 \rightarrow 1$

Code:

```
def GroupDecision(N):
    group1 = N % 5
    group2 = N % 8
    if group1 < group2:
        return(1)
    elif group2 < group1:
        return(2)
    else:
        return(0)

print(GroupDecision(N))
```

Q5) Debugging

Correct code:

```
def state(n, k):
    if k == 0:
        if (n % 4 != 0):
            print("On")
        else:
            print("Off")
    else:
        if (n % 4 != 0):
            print("Ambiguous")
        else:
            print("On")

state(n, k)
```